



ACS USNCO
U.S. National Chemistry Olympiad

2025 U.S. NATIONAL CHEMISTRY OLYMPIAD LOCAL SECTION EXAM

Prepared by the American Chemical Society Chemistry Olympiad Examinations Task Force

OLYMPIAD EXAMINATIONS TASK FORCE

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DIRECTIONS TO THE EXAMINER

This test is designed to be taken with an answer sheet on which the student records their responses. All answers are to be marked on that sheet, not written in the booklet. Each student should be provided with an answer sheet and scratch paper, both of which must be turned in with the test booklet at the end of the examination. Local Sections may use an answer sheet of their own choice.

The full examination consists of 60 multiple-choice questions representing a fairly wide range of difficulty. A periodic table and other useful information are provided on page two of this exam booklet for student reference.

Only non-programmable calculators are to be used on the ACS Local Section exam. The use of a programmable calculator, cell phone, or any other device that can access the internet or make copies or photographs during the exam is grounds for disqualification.

Suggested Time: 60 questions—110 minutes

DIRECTIONS TO THE EXAMINEE

DO NOT TURN THE PAGE UNTIL DIRECTED TO DO SO.

This is a multiple-choice examination with four choices for each question. There is only one correct or best answer to each question. When you select your choice, blacken the corresponding space on the answer sheet with your pencil. Make a heavy full mark, but no stray marks. If you decide to change your answer, be certain to erase your original answer completely.

ABBREVIATIONS AND SYMBOLS					
amount of substance	<i>n</i>	Faraday constant	<i>F</i>	molar mass	<i>M</i>
ampere	A	free energy	<i>G</i>	mole	mol
atmosphere	atm	frequency	<i>ν</i>	Planck's constant	<i>h</i>
atomic mass unit	u	gas constant	<i>R</i>	pressure	<i>P</i>
Avogadro constant	<i>N_A</i>	gram	g	rate constant	<i>k</i>
Celsius temperature	°C	hour	h	reaction quotient	<i>Q</i>
centi- prefix	c	joule	J	second	s
coulomb	C	kelvin	K	speed of light	<i>c</i>
density	d	kilo- prefix	k	temperature, K	<i>T</i>
electromotive force	<i>E</i>	liter	L	time	<i>t</i>
energy of activation	<i>E_a</i>	measure of pressure mm Hg		vapor pressure	VP
enthalpy	<i>H</i>	milli- prefix	m	volt	V
entropy	<i>S</i>	molal	<i>m</i>	volume	V
equilibrium constant	<i>K</i>	molar	M		

CONSTANTS
$R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$
$R = 0.08314 \text{ L bar mol}^{-1} \text{ K}^{-1}$
$F = 96,500 \text{ C mol}^{-1}$
$F = 96,500 \text{ J V}^{-1} \text{ mol}^{-1}$
$N_A = 6.022 \times 10^{23} \text{ mol}^{-1}$
$h = 6.626 \times 10^{-34} \text{ J s}$
$c = 2.998 \times 10^8 \text{ m s}^{-1}$
$0^\circ \text{C} = 273.15 \text{ K}$
$1 \text{ atm} = 1.013 \text{ bar} = 760 \text{ mm Hg}$
Specific heat capacity of $\text{H}_2\text{O} = 4.184 \text{ J g}^{-1} \text{ K}^{-1}$

EQUATIONS
$E = E^\circ - \frac{RT}{nF} \ln Q$ $\ln K = \left(\frac{-\Delta H^\circ}{R} \right) \left(\frac{1}{T} \right) + \text{constant}$ $\ln \left(\frac{k_2}{k_1} \right) = \frac{E_a}{R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)$

PERIODIC TABLE OF THE ELEMENTS

1 1A																	18 8A
1 H 1.008	2 2A											13 3A	14 4A	15 5A	16 6A	17 7A	2 He 4.003
3 Li 6.941	4 Be 9.012											5 B 10.81	6 C 12.01	7 N 14.01	8 O 16.00	9 F 19.00	10 Ne 20.18
11 Na 22.99	12 Mg 24.31	3 3B	4 4B	5 5B	6 6B	7 7B	8 8B	9 8B	10 8B	11 1B	12 2B	13 Al 26.98	14 Si 28.09	15 P 30.97	16 S 32.07	17 Cl 35.45	18 Ar 39.95
19 K 39.10	20 Ca 40.08	21 Sc 44.96	22 Ti 47.88	23 V 50.94	24 Cr 52.00	25 Mn 54.94	26 Fe 55.85	27 Co 58.93	28 Ni 58.69	29 Cu 63.55	30 Zn 65.39	31 Ga 69.72	32 Ge 72.61	33 As 74.92	34 Se 78.97	35 Br 79.90	36 Kr 83.80
37 Rb 85.47	38 Sr 87.62	39 Y 88.91	40 Zr 91.22	41 Nb 92.91	42 Mo 95.95	43 Tc (98)	44 Ru 101.1	45 Rh 102.9	46 Pd 106.4	47 Ag 107.9	48 Cd 112.4	49 In 114.8	50 Sn 118.7	51 Sb 121.8	52 Te 127.6	53 I 126.9	54 Xe 131.3
55 Cs 132.9	56 Ba 137.3	57 La 138.9	72 Hf 178.5	73 Ta 180.9	74 W 183.8	75 Re 186.2	76 Os 190.2	77 Ir 192.2	78 Pt 195.1	79 Au 197.0	80 Hg 200.6	81 Tl 204.4	82 Pb 207.2	83 Bi 209.0	84 Po (209)	85 At (210)	86 Rn (222)
87 Fr (223)	88 Ra (226)	89 Ac (227)	104 Rf (261)	105 Db (262)	106 Sg (263)	107 Bh (262)	108 Hs (265)	109 Mt (266)	110 Ds (281)	111 Rg (272)	112 Cn (285)	113 Nh (286)	114 Fl (289)	115 Mc (289)	116 Lv (293)	117 Ts (294)	118 Og (294)
58 Ce 140.1	59 Pr 140.9	60 Nd 144.2	61 Pm (145)	62 Sm 150.4	63 Eu 152.0	64 Gd 157.3	65 Tb 158.9	66 Dy 162.5	67 Ho 164.9	68 Er 167.3	69 Tm 168.9	70 Yb 173.0	71 Lu 175.0				
90 Th 232.0	91 Pa 231.0	92 U 238.0	93 Np (237)	94 Pu (244)	95 Am (243)	96 Cm (247)	97 Bk (247)	98 Cf (251)	99 Es (252)	100 Fm (257)	101 Md (258)	102 No (259)	103 Lr (262)				

DIRECTIONS

- When you have selected your answer to each question, blacken the corresponding space on the answer sheet using a soft, #2 pencil. Make a heavy, full mark, but no stray marks. If you decide to change an answer, erase the unwanted mark very carefully.
- There is only one correct answer to each question. Any questions for which more than one response has been blackened **will not be counted**.
- Your score is based solely on the number of questions you answer correctly. **It is to your advantage to answer every question.**

- How many nitrogen atoms are present in 486 g of N_2O_5 ?
 (A) 1.35×10^{24} (B) 2.71×10^{24}
 (C) 5.42×10^{24} (D) 1.90×10^{25}
- What is the empirical formula of a compound that is 50% S and 50% O by mass?
 (A) SO (B) S_2O
 (C) SO_2 (D) S_2O_3
- How much chlorine is required to completely consume 13.5 g of aluminum to form aluminum chloride?
 (A) 13.5 g (B) 26.6 g
 (C) 40.5 g (D) 53.2 g
- A solution of 0.500 L of 0.150 M KI is mixed with 0.250 L of 0.200 M CaI_2 . What is the concentration of iodide in the final solution?
 (A) 0.167 M (B) 0.175 M
 (C) 0.233 M (D) 0.275 M
- The only source of copper in an ore sample is the mineral malachite, $\text{Cu}_2\text{CO}_3(\text{OH})_2$ ($M = 221.1$). A 1.00 kg sample of this ore is smelted to obtain 47.0 g of metallic copper. Assuming a complete recovery of the copper, what percentage by mass of the ore is malachite?
 (A) 4.70% (B) 8.18%
 (C) 16.4% (D) 32.8%
- Which aqueous solution has the highest boiling point when 10.00 g of the substance is dissolved in 100.0 g of water?
 (A) H_2SO_4 (B) Li_2SO_4
 (C) Na_2SO_4 (D) MgSO_4
- Which compound could be used to remove Ca^{2+} from hard water?
 (A) LiCl (B) Na_2CO_3
 (C) KNO_3 (D) K_2SO_4
- Addition of silver nitrate solution to 1.0 M solutions of which compounds will result in a yellow precipitate?
 I. NaCl II. KI
 (A) I only (B) II only
 (C) Both I and II (D) Neither I nor II
- A student is cleaning a buret in preparation for a titration in which the buret will be filled with 0.100 M NaOH. What should be used for the final rinse of the buret?
 (A) Ethanol (B) Distilled water
 (C) 0.100 M HCl (D) 0.100 M NaOH
- In which solvent does solid iodine dissolve to give a purple solution?
 (A) Water (B) Ethanol
 (C) Carbon tetrachloride (D) 0.1 M aqueous KI
- Which glassware would be most suitable for preparing a 0.1500 M solution of silver nitrate?
 (A) 50 mL volumetric flask
 (B) 50 mL round-bottom flask
 (C) 50 mL Erlenmeyer flask
 (D) 50 mL graduated cylinder
- A 2.00 g sample of $\text{NaHCO}_3(\text{s})$ is heated in a 10.00 g crucible to drive off H_2O and CO_2 . The mass data in the table were collected by heating the crucible and then periodically letting it cool to room temperature and reweighing it. Which is the first measurement after which heating could have been stopped?

Measurement	Mass of crucible and solid
0	12.00 g
1	11.48 g
2	11.24 g
3	11.25 g
4	11.24 g

- (A) Measurement 1 (B) Measurement 2
 (C) Measurement 3 (D) Measurement 4

13. 1.53 g of a gas occupies a volume of 264 mL at 722 mm Hg and 22 °C. What is the molecular formula of the gas?

(A) C₂F₆ (B) C₃HF₅
(C) C₃HF₅O (D) C₃F₆O

14. A mole of which gas will have a volume closest to 22.4 L at 0 °C and 1 atm pressure?

(A) H₂ (B) CO₂
(C) SF₆ (D) Xe

15. Two identical steel cylinders at the same temperature contain equal masses of CO(g) and C₂H₂(g). Which quantities will be greater for the cylinder containing C₂H₂?

I. Pressure II. Average molecular velocity

(A) I only (B) II only
(C) Both I and II (D) Neither I nor II

16. The critical point of carbon dioxide is 31.0 °C and 72.8 atm. Which best describes the composition of a sample of pure CO₂ at 29.0 °C and 75 atm?

(A) Only gaseous CO₂ is present.
(B) Only liquid CO₂ is present.
(C) A mixture of gaseous CO₂ and liquid CO₂ is present.
(D) Only supercritical CO₂ is present.

17. The normal boiling point of ethane (C₂H₆) is –88.5 °C, while the normal boiling point of fluoromethane (CH₃F) is –78.4 °C. A difference in which type of intermolecular forces is most responsible for the higher boiling point of fluoromethane?

(A) London dispersion forces
(B) Dipole-dipole interactions
(C) Ion-dipole interactions
(D) Hydrogen bonding

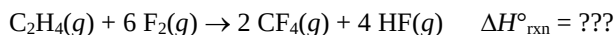
18. Cadmium metal crystallizes in a hexagonal closest packed structure. How many nearest neighbors does a cadmium atom have in this structure?

(A) 4 (B) 6 (C) 8 (D) 12

19. The standard enthalpy of fusion of 1,4-dichlorobenzene is 18.2 kJ mol^{–1} and its standard entropy of fusion is 55.3 J mol^{–1} K^{–1}. What is the melting point of 1,4-dichlorobenzene?

(A) 37.1 °C (B) 56.0 °C
(C) 101 °C (D) 329 °C

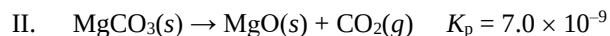
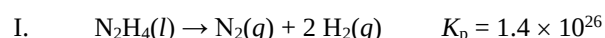
20. What is $\Delta H^\circ_{\text{rxn}}$ for the gas-phase reaction of ethene and fluorine shown below?



Substance	ΔH°_f , kJ mol ^{–1}
C ₂ H ₄ (g)	52.4
CF ₄ (g)	–679.0
HF(g)	–273.3

(A) –2503.6 kJ mol^{–1} (B) –1004.7 kJ mol^{–1}
(C) 1004.7 kJ mol^{–1} (D) 2503.6 kJ mol^{–1}

21. Which reactions are spontaneous under standard conditions?



(A) I only (B) II only
(C) Both I and II (D) Neither I nor II

22. Solid gallium melts at 29.8 °C with an enthalpy of fusion of 5.59 kJ mol^{–1}. The heat capacity of solid gallium is 26.1 J mol^{–1} K^{–1} and the heat capacity of liquid gallium is 28.5 J mol^{–1} K^{–1}. If equal masses of solid gallium at 20.0 °C and liquid gallium at 60.0 °C are mixed in a well-insulated container, what is the temperature after the mixture has attained equilibrium?

(A) 29.8 °C (B) 40.0 °C
(C) 40.4 °C (D) 40.9 °C

23. What is the K_{sp} of AgBr at 298 K?

Species	ΔG°_f (298 K), kJ mol ^{–1}
Ag ⁺ (aq)	77.1
Br [–] (aq)	–102.8
AgBr(s)	–96.9

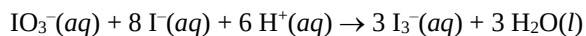
(A) 3.3×10^{-13} (B) 3.1×10^{-14}
(C) 1.0×10^{-17} (D) 3.2×10^{-22}

24. For a chemical reaction under conditions where it is at equilibrium, which statements must be true?

I. $\Delta S_{\text{rxn}} > 0$ II. $\Delta G_{\text{rxn}} = 0$

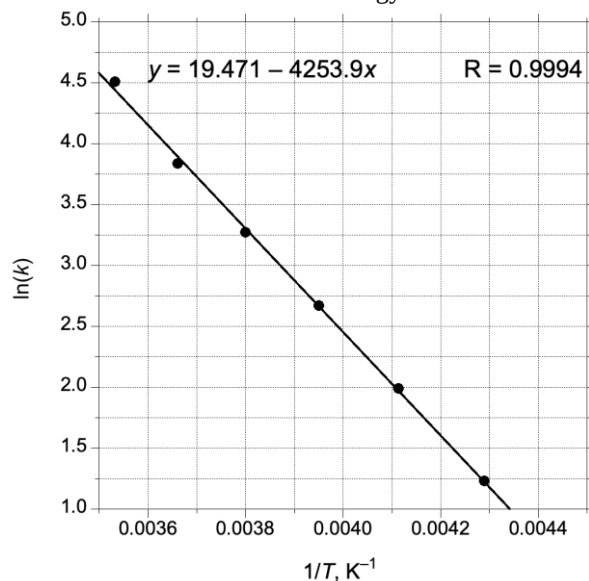
(A) I only (B) II only
(C) Both I and II (D) Neither I nor II

25. When iodate ion reacts with iodide ion in acidic solution, which statement about how concentrations are changing is correct?



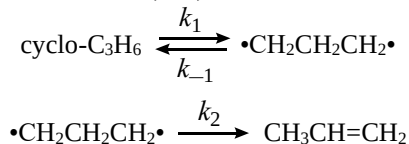
- (A) $8 \frac{\Delta[\text{I}^-]}{\Delta t} = 3 \frac{\Delta[\text{I}_3^-]}{\Delta t}$ (B) $8 \frac{\Delta[\text{I}^-]}{\Delta t} = -3 \frac{\Delta[\text{I}_3^-]}{\Delta t}$
 (C) $3 \frac{\Delta[\text{I}^-]}{\Delta t} = 8 \frac{\Delta[\text{I}_3^-]}{\Delta t}$ (D) $3 \frac{\Delta[\text{I}^-]}{\Delta t} = -8 \frac{\Delta[\text{I}_3^-]}{\Delta t}$
26. For the irreversible reaction between $\text{CO}(g)$ and $\text{Cl}_2(g)$ to form $\text{COCl}_2(g)$, the rate law is $\text{Rate} = k[\text{CO}][\text{Cl}_2]$. In an experiment where equimolar quantities of $\text{CO}(g)$ and $\text{Cl}_2(g)$ are reacted, which of the following would change linearly with reaction time?
- (A) $\ln([\text{CO}])$ (B) $\ln([\text{COCl}_2])$
 (C) $1/[\text{CO}]$ (D) $1/[\text{COCl}_2]$
27. A sample containing radium-226 was measured in 1925 to have an activity of 3.74×10^6 disintegrations per second. The same sample, re-measured in 2025, has an activity of 3.58×10^6 disintegrations per second. What is the half-life of ^{226}Ra ?
- (A) 620 years (B) 1200 years
 (C) 1600 years (D) 2300 years

28. The rate constant for the addition of pyridine to an iridium compound is measured at a variety of temperatures, and the variation of $\ln(k)$ with $1/T$ is shown below. What is the activation energy of this reaction?

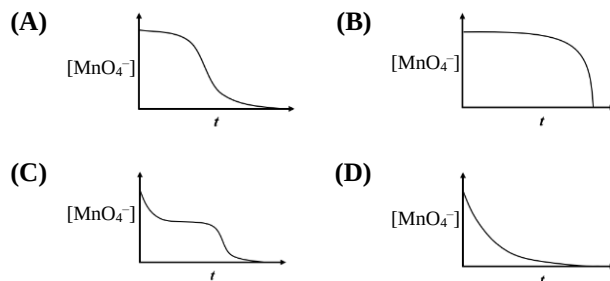


- (A) 1.82 kJ mol^{-1} (B) 4.25 kJ mol^{-1}
 (C) 35.4 kJ mol^{-1} (D) 82.8 kJ mol^{-1}

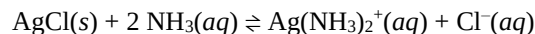
29. Cyclopropane is proposed to isomerize to propene through an intermediate diradical as shown below. The overall reaction is observed to follow the rate law $\text{Rate} = k[\text{cyclo-C}_3\text{H}_6]$. If k_2 is much smaller than k_{-1} , what is the value of k in terms of k_1 , k_{-1} , and k_2 ?



- (A) k_1 (B) k_1/k_{-1}
 (C) k_1k_2 (D) k_1k_2/k_{-1}
30. The reaction of $\text{MnO}_4^-(aq)$ and $\text{H}_2\text{C}_2\text{O}_4(aq)$ in acidic solution is autocatalytic because one of the products, $\text{Mn}^{2+}(aq)$, is a reducing agent. Starting from a mixture of reactants, which of the following graphs best shows how the concentration of $\text{MnO}_4^-(aq)$ changes over time?



31. Which statements must be correct at equilibrium?
- The rate constants for the forward and reverse reactions are equal.
 - The rates of the forward and reverse reactions are equal.
- (A) I only (B) II only
 (C) Both I and II (D) Neither I nor II
32. What is the equilibrium constant expression for the reaction below?



- (A) $K_{\text{eq}} = \frac{[\text{AgCl}][\text{NH}_3]}{[\text{Ag}(\text{NH}_3)_2^+][\text{Cl}^-]}$ (B) $K_{\text{eq}} = \frac{[\text{AgCl}][\text{NH}_3]^2}{[\text{Ag}(\text{NH}_3)_2^+][\text{Cl}^-]}$
 (C) $K_{\text{eq}} = \frac{[\text{Ag}(\text{NH}_3)_2^+][\text{Cl}^-]}{[\text{AgCl}][\text{NH}_3]^2}$ (D) $K_{\text{eq}} = \frac{[\text{Ag}(\text{NH}_3)_2^+][\text{Cl}^-]}{[\text{NH}_3]^2}$
33. Phosphoric acid has $\text{p}K_{\text{a}}$ values of $\text{p}K_{\text{a}1} = 2.15$, $\text{p}K_{\text{a}2} = 7.20$, and $\text{p}K_{\text{a}3} = 12.35$. Which species is most abundant at $\text{pH} = 11.5$?
- (A) H_3PO_4 (B) H_2PO_4^-
 (C) HPO_4^{2-} (D) PO_4^{3-}

34. A buffer is made by mixing 0.900 mol $\text{Na}[\text{CH}_3\text{CO}_2]$ and 0.500 mol HCl in enough water to make 2.50 L of solution. What is the pH of this solution? The pK_a of CH_3COOH is 4.75.
- (A) 4.49 (B) 4.65
(C) 4.85 (D) 5.01
35. A sealed container with a volume of 1.00 L contains an equilibrium mixture of $\text{N}_2\text{O}_4(\text{g})$ and $\text{NO}_2(\text{g})$ at a given temperature. Which quantities will increase if the volume of the container is increased to 2.00 L while maintaining the same temperature?
- I. The number of moles of $\text{NO}_2(\text{g})$
II. The partial pressure of $\text{NO}_2(\text{g})$
- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II
36. A solution is 0.05 M in each of the anions fluoride (F^-), carbonate (CO_3^{2-}), and phosphate (PO_4^{3-}). As calcium ion is added slowly to this solution, in what order will the solids precipitate? The K_{sp} values are: CaF_2 , 3.9×10^{-11} ; CaCO_3 , 3.8×10^{-9} ; $\text{Ca}_3(\text{PO}_4)_2$, 1.0×10^{-25} .
- (A) First CaF_2 , then CaCO_3 , then $\text{Ca}_3(\text{PO}_4)_2$
(B) First CaF_2 , then $\text{Ca}_3(\text{PO}_4)_2$, then CaCO_3
(C) First $\text{Ca}_3(\text{PO}_4)_2$, then CaF_2 , then CaCO_3
(D) First $\text{Ca}_3(\text{PO}_4)_2$, then CaCO_3 , then CaF_2
37. What is the average oxidation state of iron in halotrichite, $\text{FeAl}_2(\text{SO}_4)_4 \cdot 22 \text{H}_2\text{O}$?
- (A) 0 (B) +2 (C) +2.5 (D) +3
38. Which half-reaction takes place at the anode during electrolysis of aqueous sodium iodide?
- (A) $\text{Na}^+(\text{aq}) + \text{e}^- \rightarrow \text{Na}(\text{s})$
(B) $2 \text{H}_2\text{O}(\text{l}) + 2 \text{e}^- \rightarrow \text{H}_2(\text{g}) + 2 \text{OH}^-(\text{aq})$
(C) $2 \text{I}^-(\text{aq}) \rightarrow \text{I}_2(\text{s}) + 2 \text{e}^-$
(D) $2 \text{H}_2\text{O}(\text{l}) \rightarrow \text{O}_2(\text{g}) + 4 \text{H}^+(\text{aq}) + 4 \text{e}^-$
39. In an electrochemical cell, the reaction shown below occurs. Which changes will decrease the cell potential?
- $$\text{Fe}^{2+}(\text{aq}) + \text{Ag}^+(\text{aq}) \rightarrow \text{Fe}^{3+}(\text{aq}) + \text{Ag}(\text{s})$$
- I. Decreasing the concentration of $\text{Fe}^{2+}(\text{aq})$
II. Increasing the size of the silver electrode
- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II
40. 180 mL of a solution that is 0.024 M in both AgNO_3 and $\text{Cu}(\text{NO}_3)_2$ is electrolyzed with a current of 0.70 A for 350 s. What deposits on the cathode?
- (A) Ag only
(B) Cu only
(C) 2.5×10^{-3} mol Ag and some Cu
(D) 2.5×10^{-3} mol Cu and some Ag
41. What is the standard potential of a fuel cell where methanol is oxidized to carbon dioxide and oxygen is reduced to water?
- | Half-reaction | E° , V |
|--|---------------|
| $\text{CO}_2(\text{g}) + 6 \text{H}^+(\text{aq}) + 6 \text{e}^- \rightarrow \text{CH}_3\text{OH}(\text{l}) + \text{H}_2\text{O}(\text{l})$ | -0.02 |
| $\text{O}_2(\text{g}) + 4 \text{H}^+(\text{aq}) + 4 \text{e}^- \rightarrow 2 \text{H}_2\text{O}(\text{l})$ | 1.23 |
- (A) 0.30 V (B) 0.31 V
(C) 1.21 V (D) 1.25 V
42. The half-cell potential at 298 K for the reduction of an oxoanion becomes more negative by 0.118 V for each increase of 1 pH unit. How many protons and electrons are involved in this half-reaction?
- (A) 1 proton, 1 electron (B) 1 proton, 2 electrons
(C) 2 protons, 1 electron (D) 3 protons, 2 electrons
43. Which atom has the smallest first ionization energy?
- (A) Mg (B) Al (C) Si (D) P
44. How many electrons in a ground-state atom of uranium (U) have the principal quantum number $n = 4$?
- (A) 4 (B) 16 (C) 18 (D) 32
45. Which property of a photon is directly proportional to its frequency?
- (A) Energy (B) Momentum
(C) Velocity (D) Wavelength
46. Photoelectron spectroscopy of Ar provides information about which properties of Ar?
- I. Ionization energy II. Electronic subshell structure
- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II

47. Which statements about fluorine are correct?
I. It has the highest electronegativity of any element.
II. Its electron affinity is the largest in magnitude of any element.

(A) I only (B) II only
(C) Both I and II (D) Neither I nor II

48. Thorium-232 decays by emission first of an alpha particle, then of a beta particle, then of a second beta particle, and then of a second alpha particle. What is the daughter nucleus after the fourth disintegration?

(A) ^{228}Th (B) ^{228}Ac
(C) ^{224}Ra (D) ^{224}Rn

49. Which molecule is trigonal planar?

(A) NH_3 (B) NF_3 (C) SO_3 (D) ClF_3

50. Which molecule has the shortest N–O bond?

(A) NO (B) NO_2 (C) N_2O (D) N_2O_4

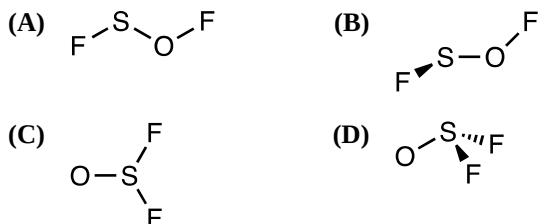
51. How many lone pairs are there in the Lewis structure of nitromethane, CH_3NO_2 ?

(A) 3 (B) 4 (C) 5 (D) 7

52. Which is the strongest acid in aqueous solution?

(A) NH_4^+ (B) H_2O (C) PH_4^+ (D) H_2S

53. Which is the best representation of the three-dimensional arrangement of the atoms in thionyl fluoride, SOF_2 ?



54. Which ion is diamagnetic?

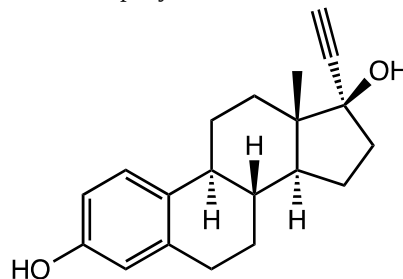
(A) $\text{Fe}(\text{CN})_6^{4-}$ (B) CoCl_4^{2-}
(C) $\text{Ni}(\text{H}_2\text{O})_6^{2+}$ (D) $\text{Cu}(\text{NH}_3)_4^{2+}$

55. What is the relationship between the two alcohols?



(A) Structural isomers (B) Enantiomers
(C) Diastereomers (D) Identical

56. In ethinylestradiol, shown below, how many sp^2 -hybridized and sp -hybridized carbons are there?



(A) 6 sp^2 , 2 sp (B) 5 sp^2 , 8 sp
(C) 11 sp^2 , 8 sp (D) 12 sp^2 , 8 sp

57. Which statement most accurately describes the most stable conformation of cyclohexane?

(A) All six carbon atoms are in the same plane.
(B) All bond angles are 109.5° .
(C) All hydrogens are eclipsed relative to the hydrogen atoms on adjoining carbons.
(D) The carbon-carbon bonds alternate between short and long distances.

58. The tertiary alcohol 2-methyl-2-propanol reacts with concentrated aqueous hydrochloric acid at 0°C to form 2-chloro-2-methylpropane. How is the mechanism of this reaction best described?

(A) $\text{S}_{\text{N}}1$ (B) $\text{S}_{\text{N}}2$ (C) $\text{E}1$ (D) $\text{E}2$

59. Which statement most accurately describes the properties of ketones and esters?

(A) Ketones are weak acids; esters are strong acids.
(B) Ketones react by nucleophilic addition; esters react by nucleophilic substitution.
(C) Ketones are readily reduced; esters are readily oxidized.
(D) The carbonyl group of a ketone is resonance stabilized; the carbonyl group of an ester is not resonance stabilized.

60. Which is not found in a typical nucleic acid?

(A) Carboxylic acid (B) Dialkyl phosphate
(C) Heterocyclic base (D) Five-carbon sugar

END OF TEST

Olympiad 2025
USNCO Local Section Exam
KEY

Number Answer		Number Answer	
1.	C	31.	B
2.	C	32.	D
3.	D	33.	C
4.	C	34.	B
5.	B	35.	A
6.	B	36.	B
7.	B	37.	B
8.	B	38.	C
9.	D	39.	A
10.	C	40.	A
11.	A	41.	D
12.	C	42.	C
13.	C	43.	B
14.	A	44.	D
15.	C	45.	A
16.	B	46.	C
17.	B	47.	A
18.	D	48.	C
19.	B	49.	C
20.	A	50.	A
21.	A	51.	C
22.	A	52.	C
23.	A	53.	D
24.	B	54.	A
25.	D	55.	D
26.	C	56.	A
27.	C	57.	B
28.	C	58.	A
29.	D	59.	B
30.	A	60.	A